Polygons & Quadrilaterals

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In ΔABC studying proving triangles similar, $m\angle A = 38^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 90°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (38^\circ + 52^\circ) = 90^\circ$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

In ΔABC studying applications of similar triangles, $m\angle A = 39^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 96°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (39^\circ + 45^\circ) = 96^\circ$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Find the area of a circle with radius 5.

Apply the circle area formula $A = \pi r^2$: $A = \pi(5)^2 = \mathbf{25\pi} \approx 78.54$ Trap Warning: Do NOT confuse area (πr^2) with circumference ($2\pi r = 10\pi$). Most ACT answer choices leave π exact.

4 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

A solid metal cone with base radius 4 cm and height 9 cm is submerged into a cylindrical beaker with internal radius 6 cm partially filled with water. Assuming the cone is fully submerged and no water overflows, by how many centimeters does the water level in the beaker rise?

Step 1: Compute the volume of the submerged cone: $V_{\text{cone}} = \frac{1}{3}\pi r_{\text{cone}}^2 h_{\text{cone}} = \frac{1}{3}\pi(4^2)(9) = \frac{1}{3}\pi(16)(9) = 48\pi$ cm^3. Step 2: The volume of water displaced forms a cylinder with radius $R = 6$ cm and height Δh: $V_{\text{displaced}} = \pi R^2(\Delta h) = \pi(6^2)\Delta h = 36\pi\Delta h$. Step 3: Equate volumes and solve for Δh.

5 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Let $f(x) = \frac{2x-7}{5}$. (a) Find $f^{-1}(x)$. (b) Solve the equation $f(f^{-1}(x)) = 9$. (c) Find the value of k for which $f(k) = f^{-1}(k)$.

(a) Find $f^{-1}(x)$: Let $y = \frac{2x-7}{5}$. Swap x and y: $x = \frac{2y-7}{5} \Rightarrow 5x = 2y - 7 \Rightarrow 2y = 5x + 7 \Rightarrow \mathbf{f^{-1}(x) = \frac{5x+7}{2}}$. (b) Composition identity: By definition, for all x in the domain of f^{-1}, $f(f^{-1}(x)) = x$. Therefore, $f(f^{-1}(x)) = 9 \Rightarrow \mathbf{x = 9}$. (c) Point on line $y = x$: If $f(k) = f^{-1}(k)$, the intersection must lie on the line of reflection $y = x$. Setting $f(k) = k$.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find all real solutions to the equation $|2x - 7| = 3x - 1$.

Step 1: Split into two cases based on the definition of absolute value: Case 1: $2x - 7 = 3x - 1 \Rightarrow -7 + 1 = 3x - 2x \Rightarrow x = -6$. Check Case 1: $|2(-6) - 7| = |-19| = 19$, but the right side is $3(-6) - 1 = -19$. Since $19 eq -19$, $x = -6$ is an extraneous solution. Case 2: $2x - 7 = -(3x - 1) = -3x + 1 \Rightarrow 2x + 3x = 1 + 7 \Rightarrow 5x = 8 \Rightarrow x = \frac{8}{5} = 1.6$. Check Case 2: $|2(1.6) - 7| = |3.2 - 7| = |-3.8| = 3.8$. Right side: $3(1.6) - 1 = 4.8 - 1 = 3.8$ (Valid ✓). The o.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A parabola has its vertex at $(3, -4)$ and passes through the point $(1, 0)$. (a) Write the equation of the parabola in standard form $y = ax^2 + bx + c$. (b) Determine the coordinates of the other x-intercept.

(a) Use vertex form: With vertex $(h, k) = (3, -4)$: $y = a(x - 3)^2 - 4$ Substitute $(x, y) = (1, 0)$ to find a: $0 = a(1 - 3)^2 - 4 \Rightarrow 0 = 4a - 4 \Rightarrow 4a = 4 \Rightarrow a = 1$. Expand to standard form: $y = 1(x^2 - 6x + 9) - 4 = \mathbf{x^2 - 6x + 5}$. (b) Find the other x-intercept: By symmetry, the x-intercepts are equidistant from the axis of symmetry $x = 3$. Since one intercept is at $x = 1$ (distance $3 - 1 = 2$), the other intercept is at $x = 3 + 2 = 5$, giving **(5, 0)**. (V.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the ordered dataset of 12 numbers: 14, 16, 18, 20, 22, 24, 26, 28, 30, 34, 38, 72. (a) Determine Q_1, the median, Q_3, and the IQR. (b) Use Tukey's $1.5 \times$ IQR rule to determine whether 72 is an outlier.

(a) Five-Number Summary Calculations:Lower half (first 6 values): 14, 16, 18, 20, 22, 24 $\Rightarrow Q_1 = \frac{18 + 20}{2} = \mathbf{19}$.Median of all 12 values: $\frac{24 + 26}{2} = \mathbf{25}$.Upper half (last 6 values): 26, 28, 30, 34, 38, 72 $\Rightarrow Q_3 = \frac{30 + 34}{2} = \mathbf{32}$.IQR $= Q_3 - Q_1 = 32 - 19 = \mathbf{13}$. (b) Outlier Boundary Check: Upper Outlier Boundary $= Q_3 + 1.5 \times \text{IQR} = 32 + 1.5(13) = 32 + 19.5 = \mathbf{51.5}$. Since $72 > 51.5$, the va.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

An equilateral triangle has a side length of 12 cm. Find: (a) the exact length of its altitude, and (b) its exact area.

(a) Altitude using $30^\circ - 60^\circ - 90^\circ$ triangle: Dropping an altitude splits the equilateral triangle into two congruent $30^\circ - 60^\circ - 90^\circ$ right triangles with short leg $b/2 = 6$ cm and hypotenuse 12 cm. The altitude is opposite the 60° angle: $h = 6\sqrt{3}$ cm. (b) Area formula: $A = \frac{1}{2}bh = \frac{1}{2}(12)(6\sqrt{3}) = \mathbf{36\sqrt{3}}$ cm². (Alternatively.

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10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Two assembly line components operate independently. Component A has a 95% reliability rate (probability of functioning properly is 0.95), and Component B has a 90% reliability rate (probability of functioning is 0.90). What is the probability that at least one of the two components fails?

Step 1: Use the complement rule. The event "at least one fails" is the exact complement of "both components function properly": $P(\geq 1 \text{ fails}) = 1 - P(\text{both function})$. Step 2: Since the components operate independently: $P(\text{both function}) = P(A) \times P(B) = 0.95 \times 0.90 = 0.855$. Step 3: Calculate the complement: $P(\geq 1 \text{ fails}) = 1 - 0.855 = \mathbf{0.145\ (14.5\%)}$. The probability that at least one component fails is 0.145 (or 14.5%).

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

For what value of the constant a will the system of equations $ax + 4y = 7$ and $3x + 2y = 5$ have no solution?

Step 1: A linear system has no solution when the lines are parallel and distinct (slopes are equal, y-intercepts differ). In standard form, this requires: $\frac{a}{3} = \frac{4}{2} eq \frac{7}{5}$ Step 2: Solve the slope equality: $\frac{a}{3} = 2 \Rightarrow a = 2 \times 3 = \mathbf{6}$ Check: When $a = 6$, the ratio of coefficients is $\frac{6}{3} = \frac{4}{2} = 2$, while the constant ratio is $\frac{7}{5} = 1.4 eq 2$. Therefore, the lines are strictly parallel with no solution.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Determine the maximal real domain of the function $f(x) = \frac{\sqrt{3x-12}}{x^2-25}$. Express your answer in interval notation.

Step 1: For the square root in the numerator to be defined over the reals, the radicand must be non-negative: $3x - 12 \geq 0 \Rightarrow 3x \geq 12 \Rightarrow x \geq 4$. Step 2: For the denominator to be non-zero: $x^2 - 25 eq 0 \Rightarrow x eq 5 \quad \text{and} \quad x eq -5$. Step 3: Intersect the conditions $x \geq 4$ and $x eq 5$ (note that $x = -5$ is already outside $x \geq 4$): $[4, 5) \cup (5, \infty)$. The maximal real domain is **$[4, 5) \cup (5, \infty)$**.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find the equation of the perpendicular bisector of the line segment joining $P(2, -1)$ and $Q(8, 7)$. Give your answer in the form $y = mx + c$.

Step 1: Find the midpoint M of PQ: $M = \left(\frac{2+8}{2}, \frac{-1+7}{2}\right) = (5, 3)$ Step 2: Find the gradient of line PQ: $m_{PQ} = \frac{7-(-1)}{8-2} = \frac{8}{6} = \frac{4}{3}$ Step 3: Determine the perpendicular gradient: $m_{\perp} = -\frac{3}{4}$ Step 4: Form the equation passing through $M(5, 3)$: $y - 3 = -\frac{3}{4}(x - 5) \Rightarrow y - 3 = -\frac{3}{4}x + \frac{15}{4}$
$y = -\frac{3}{4}x + \frac{27}{4} \quad \text{or} \quad y = -0.75x + 6.75$

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14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

If $x - 2$ is a factor of the polynomial $P(x) = x^3 - kx^2 + 3x - 10$, find the value of k, and determine all real roots of $P(x)$.

Step 1: By the Factor Theorem, $x - 2$ is a factor $\Leftrightarrow P(2) = 0$: $P(2) = (2)^3 - k(2)^2 + 3(2) - 10 = 0 \Rightarrow 8 - 4k + 6 - 10 = 0 \Rightarrow 4 - 4k = 0 \Rightarrow \mathbf{k = 1}$. Step 2: Substitute $k = 1$ into $P(x) = x^3 - x^2 + 3x - 10$ and divide by $(x - 2)$ using polynomial long division or synthetic division: $x^3 - x^2 + 3x - 10 = (x - 2)(x^2 + x + 5)$. Step 3: Solve $x^2 + x + 5 = 0$. Its discriminant is \(\Delta = (1)^2 - 4(1)(5) = 1 - 20 = -19

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 10.01 Coordinate Geometry**

A rectangular water storage reservoir has dimensions 2.5 meters by 3 meters by 1.2 meters. A drainage pump empties the reservoir at a constant rate of 75 liters per minute. Given that $1\ \text{m}^3 = 1000\ \text{liters}$, how many hours will it take to completely empty the full reservoir?

Step 1: Find the reservoir volume in cubic meters: $V = 2.5 \times 3.0 \times 1.2 = 9.0\ \text{m}^3$. Step 2: Convert volume to liters: $V = 9.0\ \text{m}^3 \times 1000\ \text{L/m}^3 = 9000\ \text{liters}$. Step 3: Compute drainage time in minutes: $t_{\text{min}} = \frac{9000\ \text{L}}{75\ \text{L/min}} = 120\ \text{minutes}$. Step 4: Convert to hours: $t = \frac{120\ \text{min}}{60\ \text{min/h}} = \mathbf{2}\ \text{hours}$. It takes exactly 2 hours.

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 11.01 Probability**

In the system $ax + 4y = 18$ and $3x + 2y = 9$, the system has no solution. What is the value of a?

A system of linear equations has no solution if the lines are parallel (equal slopes) with different y-intercepts. Equal slopes require the ratio of the x-coefficients to equal the ratio of the y-coefficients: $\frac{a}{3} = \frac{4}{2} = 2$. Multiplying both sides by 3 gives $a = 6$. (Since $\frac{18}{9} = 2$, note that the constants would produce identical lines if $a = 6$; however, to make lines parallel and distinct with non-overlapping constraints, $a = 6$).

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 2.01 Lines & Angles**

In triangle OAB, $\vec{OA} = 3\mathbf{a}$ and $\vec{OB} = 6\mathbf{b}$. Point P lies on AB such that $AP : PB = 1 : 2$. Find $\vec{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ in its simplest form.

Step 1: Express $\vec{AB}$: $\vec{AB} = \vec{AO} + \vec{OB} = -3\mathbf{a} + 6\mathbf{b}$ Step 2: Find $\vec{AP}$: $\vec{AP} = \frac{1}{3}\vec{AB} = \frac{1}{3}(-3\mathbf{a} + 6\mathbf{b}) = -\mathbf{a} + 2\mathbf{b}$ Step 3: Find $\vec{OP}$: $\vec{OP} = \vec{OA} + \vec{AP} = 3\mathbf{a} + (-\mathbf{a} + 2\mathbf{b}) = 2\mathbf{a} + 2\mathbf{b}$

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18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Triangles & Congruence

A line segment connects points $P(-3, 7)$ and $Q(5, -1)$. Determine: (a) the exact length of PQ, (b) the coordinates of the midpoint M, and (c) the slope of the perpendicular bisector of PQ.

(a) Distance formula: $PQ = \sqrt{(5-(-3))^2 + (-1-7)^2} = \sqrt{8^2 + (-8)^2} = \sqrt{64+64} = \sqrt{128} = \mathbf{8\sqrt{2}}$. (b) Midpoint formula: $M = \left(\frac{-3+5}{2}, \frac{7+(-1)}{2}\right) = \left(\frac{2}{2}, \frac{6}{2}\right) = \mathbf{(1, 3)}$. (c) Perpendicular slope: Slope of PQ is $m = \frac{-1-7}{5-(-3)} = \frac{-8}{8} = -1$. The perpendicular bisector has negative reciprocal slope: $m_{\perp} = -\frac{1}{-1} = \mathbf{+1}$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

The graph of $y = f(x)$ passes through the points $(-2, 8)$ and $(4, -6)$. If $g(x) = 3f(x-5) + 2$, determine the corresponding points on the graph of $g(x)$.

Transformation Mapping: For any point (x, y) on f, the corresponding point on $g(x) = 3f(x-5) + 2$ has coordinates: $(x', y') = (x+5, 3y+2)$. For point 1: $(-2, 8)$: $x'_1 = -2+5 = 3, \quad y'_1 = 3(8)+2 = 24+2 = 26 \Rightarrow \mathbf{(3, 26)}$. For point 2: $(4, -6)$: $x'_2 = 4+5 = 9, \quad y'_2 = 3(-6)+2 = -18+2 = -16 \Rightarrow \mathbf{(9, -16)}$. The corresponding points on $g(x)$ are **$(3, 26)$** and **$(9, -16)$**.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

Solve the quadratic equation $3x^2 + 10x - 8 = 0$ by factoring.

Step 1: Find two integers whose product is $a \cdot c = 3(-8) = -24$ and whose sum is $b = 10$. The factors are $+12$ and -2. Step 2: Split the middle term and factor by grouping: $3x^2 + 12x - 2x - 8 = 0 \Rightarrow 3x(x+4) - 2(x+4) = 0 \Rightarrow (3x-2)(x+4) = 0$ Step 3: Set each factor to zero: $3x - 2 = 0 \Rightarrow x = \frac{2}{3}, \quad x + 4 = 0 \Rightarrow x = -4$ The solutions are **$x = \frac{2}{3}$** and **$x = -4$**.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying review: similar figures, $m\angle A = 36^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 99° ✓

b. 109°

c. 91°

d. 144°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (36^\circ + 45^\circ) = 99^\circ$.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying review: identifying similar triangles, $m\angle A = 37^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 98° ✓

b. 108°

c. 90°

d. 143°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (37^\circ + 45^\circ) = 98^\circ$.

23 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In parallelogram $ABCD$, $\angle A = (3x + 4)^\circ$ and $\angle B = (5x + 16)^\circ$. Find $\angle A$.

a. 116°

b. **64° ✓**

c. 74°

d. 20°

Consecutive angles are supplementary: $(3x + 4) + (5x + 16) = 180$. Solving gives $x = 20$, so $\angle A = 64^\circ$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In parallelogram $ABCD$, $\angle A = 132^\circ$. What is the measure of $\angle B$?

a. 228°

b. 132°

c. **48° ✓**

d. 42°

Consecutive angles in a parallelogram are supplementary: $\angle B = 180 - 132 = 48^\circ$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the measure of one exterior angle of a regular 30-gon?

a. **12° ✓**

b. 348°

c. 168°

d. 24°

Exterior angles of a regular polygon sum to 360°: each $= 360/30 = 12^\circ$.

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The diagonals of a parallelogram bisect each other. If one diagonal has length 20, what is the length of half of that diagonal?

a. 10 ✓

b. 20

c. 12

d. 40

Diagonals of a parallelogram bisect each other, so half $= 20/2 = 10$.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

How many diagonals does a convex polygon with 10 sides have?

a. 35 ✓

b. 34

c. 10

d. 45

Number of diagonals $= \frac{n(n-3)}{2} = \frac{10(10-3)}{2} = 35$.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A trapezoid has parallel bases of length 28 and 28. What is the length of the midsegment?

a. 28 ✓

b. 56

c. 31

d. 0

Midsegment $= \frac{b_1 + b_2}{2} = \frac{28 + 28}{2} = 28$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In isosceles trapezoid $ABCD$, base AB is parallel to base CD with $AB > CD$. The midsegment MN connects the midpoint M of leg AD to the midpoint N of leg BC. Diagonals AC and BD intersect MN at points P and Q, respectively. If $MN = 16$, $PQ = 5$, and the leg length is $AD = 13$, what is the length of diagonal AC?

a. **20** ✓

b. 18

c. $5\sqrt{17}$

d. $2\sqrt{61}$

We solve the problem in three geometric steps using midsegment and isosceles trapezoid properties: 1. **Determine the Base Lengths:** By the Trapezoid Midsegment Theorem, the midsegment MN has length: $MN = \frac{AB + CD}{2} = 16 \Rightarrow AB + CD = 32$ In ΔACD, MP connects the midpoint of AD parallel to base CD, so $MP = \frac{1}{2}CD$. In ΔABD, MQ connects the midpoint of AD parallel to base AB, so $MQ = \frac{1}{2}AB$. Therefore, the segment along the midsegment between the two.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In rhombus $ABCD$, the diagonals AC and BD intersect at point E. If $m\angle BAC = (3x + 10)^\circ$ and $m\angle ABD = (2x - 5)^\circ$, what is the measure of $\angle ABC$?

a. 29°

b. 58° ✓

c. 61°

d. 122°

1. In any rhombus, the diagonals are perpendicular to each other. Therefore, $AC \perp BD$, which means $m\angle AEB = 90^\circ$. 2. In right triangle ΔABE, the acute angles $\angle EAB$ (which is $\angle BAC$) and $\angle EBA$ (which is $\angle ABD$) are complementary: $(3x + 10) + (2x - 5) = 90$ $5x + 5 = 90 \Rightarrow 5x = 85 \Rightarrow x = 17$ 3. Calculate $m\angle ABD$: $m\angle ABD = 2(17) - 5 = 34 - 5 = 29^\circ$ 4. The diagonals of a rhombus bisect the vertex angles. Thus, diagonal BD bisects.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Four interior angles of a convex pentagon have measures of 110°, 120°, 95°, and 130°. What is the measure of the fifth interior angle?

a. 75°

b. 85° ✓

c. 95°

d. 105°

The sum of the interior angle measures of an n-sided convex polygon is given by $(n-2)\times 180^\circ$. For a pentagon ($n=5$): $(5-2)\times 180^\circ = 3\times 180^\circ = 540^\circ$ Sum the measures of the four given interior angles: $110^\circ + 120^\circ + 95^\circ + 130^\circ = 455^\circ$ Subtract this sum from the total interior angle sum to find the measure of the fifth angle: $540^\circ - 455^\circ = 85^\circ$ Common misconceptions: - Using the formula for a quadrilateral (360°) or miscal.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In the rhombus ABCD shown below, the diagonals AC and BD intersect at point E. If the length of diagonal AC is $10x$ units and the length of diagonal BD is $12x$ units, and the perimeter of the rhombus is 52 units, what is the length of side AB?

a. 13 **units** ✓

b. 26 units

c. 10 units

d. 12 units

A rhombus is a quadrilateral with all four sides equal in length. Its diagonals bisect each other at right angles. Let the side length of the rhombus be 's'. The perimeter of the rhombus is given by 4s. We are given that the perimeter is 52 units, so $4s = 52$. Dividing by 4, we get $s = 13$ units. Therefore, the length of side AB is 13 units. The lengths of the diagonals are not needed to find the side length when the perimeter is given.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A regular hexagon has interior angles that are all equal. What is the measure of each interior angle of a regular hexagon?

a. 100°

b. 108°

c. **120°** ✓

d. 150°

The sum of the interior angles of a convex n-gon is given by the formula $(n-2)\times 180^\circ$. For a hexagon, $n=6$. Thus, the sum of the interior angles is $(6-2)\times 180^\circ = 4\times 180^\circ = 720^\circ$. Since a regular hexagon has 6 equal interior angles, each interior angle measures $\frac{720^\circ}{6} = 120^\circ$.

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34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

What is the sum of the interior angles of a polygon with 25 sides?

a. 4320°

b. 4500°

c. 3960°

d. **4140° ✓**

Sum of interior angles $= (n-2) \times 180 = (25-2) \times 180 = 4140^\circ$.

35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Two similar polygons have corresponding sides in the ratio $3:8$. What is the ratio of their areas?

a. 3 : 8

b. 6 : 16

c. **9 : 64 ✓**

d. 27 : 512

The ratio of areas of similar figures equals the square of the ratio of corresponding side lengths: $\left(\frac{3}{8}\right)^2 = \frac{9}{64}$, so the ratio of their areas is $9:64$.

36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

In a regular polygon, each interior angle is 1 times its corresponding exterior angle. How many sides does the polygon have?

a. 3

b. 8

c. 6

d. **4 ✓**

Interior $= k \cdot$ Exterior and Interior+Exterior $= 180 \Rightarrow$ Exterior $= \frac{180}{k+1}$, $n = \frac{360}{\text{Exterior}} = 2(k+1) = 4$.

37 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A rectangle has length 3 and width 4. Find the length of its diagonal.

a. 13

b. 12

c. 7

d. **5 ✓**

Applying Quadrilaterals principles, $d = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$. Therefore, the verified correct option is 5.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Triangles & Congruence

A rectangle has diagonal 5 and length 3. Find the width.

a. 6

b. 8

c. 2

d. **4** ✓

Applying Quadrilaterals principles, $w = \sqrt{5^2 - 3^2} = \sqrt{16} = 4$. Therefore, the verified correct option is 4.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

A trapezoid has parallel sides 6 and 10 with height 4. Find its area.

a. 35

b. 24

c. **32** ✓

d. 64

Applying Quadrilaterals principles, $A = \frac{1}{2}(6 + 10)(4) = 32$. Therefore, the verified correct option is 32.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

A rectangle has perimeter 38 and length 10. Find the width.

a. 28

b. 19

c. **9** ✓

d. 11

Applying Quadrilaterals principles, $w = \frac{38}{2} - 10 = 9$. Therefore, the verified correct option is 9.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Circles

A parallelogram has base 9 and height 5. Find its area.

a. 28

b. 14

c. 54

d. **45** ✓

Applying Quadrilaterals principles, $A = bh = 9 \times 5 = 45$. Therefore, the verified correct option is 45.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Area, Surface Area & Volume

The sum of interior angles of a convex polygon is 360°. How many sides does it have?

a. 7

b. 5

c. **4** ✓

d. 2

Applying Polygons principles, $n = \frac{360}{180} + 2 = 4$. Therefore, the verified correct option is 4.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 9.01 Transformations & Symmetry

A rectangle has area 48 and one side 6. Find the other side.

a. 42

b. 6

c. 10

d. **8** ✓

Applying Quadrilaterals principles, $\frac{48}{6} = 8$. Therefore, the verified correct option is 8.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Foundations & Reasoning

A rectangle has length 9 and width 4. Find its perimeter.

a. 13

b. 28

c. 36

d. **26** ✓

Applying Quadrilaterals principles, $P = 2(9 + 4) = 26$. Therefore, the verified correct option is 26.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Coordinate Geometry

A rhombus has side length 6. Find its perimeter.

a. 36

b. 28

c. **24** ✓

d. 12

Applying Quadrilaterals principles, $P = 4(6) = 24$. Therefore, the verified correct option is 24.

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